



ADAPTIVE EDGE ELEMENT METHOD FOR A QUASILINEAR PROBLEM IN ELECTROMAGNETISM WITH STRONG CONVERGENCE FOR GAUSS' LAW

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This paper is concerned with adaptive edge element methods based on a regularized formulation([1]-[3]) for a quasilinear $H(\text{curl})$ -elliptic problem from electromagnetism. The lowest order 1st-kind and 2nd-kind Nédélec elements are employed in discrete problems. Our theoretical results include an $H(\text{curl})$ -strong convergence of the sequence of discrete solutions by the proposed adaptive algorithm towards the exact solution and a vanishing limit of the sequence of relevant error estimators. Moreover, the adaptive algorithm is shown to ensure a strong convergence for the associated discrete Gauss' law in the $H\{-1\}$ -norm. Finally, with the Dörfler marking included, the adaptive algorithm is shown to be contractive for the sum of an $H(\text{curl})$ -error related quantity and a scaled estimator up to a high order term between two consecutive adaptive loops.

References

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